

MADHUKAR AI

Modern Analytics and Data Hub for User-Friendly Karnataka Anti-Crime Response Platform

Complete Technical & Non-Technical Architecture, Feature Workflows, Data Pipeline & System Specifications

Document Version:	1.0 (Production Candidate)
Target Event:	KSP Datathon 2026 Grand Finale — Bangalore (Sept 18, 2026)
Classification:	RESTRICTED // INTERNAL DEV TEAM ONLY
Prepared By:	MADHUKAR AI Core Engineering Team
Cloud Platform:	Zoho Catalyst (Serverless + DataStore + QuickML GLM-4.7)
Frontend Framework:	React 19 SPA + Leaflet GIS + Recharts + OTPAuth

NOTICE: This specification outlines the end-to-end architecture, backend microservices, mathematical security models, and frontend interfaces for the prototype presented in the Top 10 Finals. Do not publish API secrets, TOTP keys, or live police datasets.

1. Executive Summary & System Overview

Non-Technical Perspective (For Police Leadership & Jury):

Modern police forces generate thousands of First Information Reports (FIRs) every month across dozens of police stations. Traditionally, identifying criminal patterns across district boundaries requires manual intelligence coordination, phone calls, and physical register cross-referencing. **MADHUKAR AI** transforms this workflow into an autonomous, real-time command center. An Investigating Officer (IO) or Police Commissioner can open the platform, view citywide crime severity heatmaps, immediately see repeat offender linkages across different police stations, and ask complex investigation questions in plain English or Kannada — receiving instant, forensically accurate intelligence answers.

Technical Perspective (For Software Architects & Evaluators):

MADHUKAR AI is a cloud-native, serverless intelligence architecture hosted on **Zoho Catalyst**. It features an in-process high-speed data cache (dataCache.js) that unifies 26 relational DataStore tables into pre-joined operational views. The intelligence layer integrates **GLM-4.7-Flash** (a 30-Billion parameter Mixture-of-Experts neural model via Catalyst QuickML) running an automated tool-calling loop that translates natural language queries into deterministic database queries. Security is governed by an enterprise-grade Zero-Trust model featuring **hardware-backed per-officer TOTP 2FA** (AES-256-GCM encrypted secrets stored in DataStore with QR onboarding) protecting all Personally Identifiable Information (PII) under the Digital Personal Data Protection (DPDP) Act standards.

Pillar	Operational Impact	Technical Mechanism
Predictive Hotspots	Enables preventative patrol allocation before crimes occur.	Kernel density estimation + 30-day rolling severity forecasts.
Forensic MO Linkage	Detects inter-district serial offenders automatically.	Statistical scoring engine (0-100) with cross-category gate.
MADHUKAR AI Copilot	Zero-training conversational investigation assistant.	GLM-4.7-Flash MoE LLM with automated ReAct tool-calling.
Zero-Trust PII 2FA	Protects victim & witness identities with strict audit trails.	Per-user AES-256 TOTP QR setup + self-service email OTP recovery.

2. Deep Tech Stack Specifications

Layer	Technology & Version	Purpose & Architectural Rationale
Frontend Framework	React 19.2.7 + React Router 7.18.1	Modern functional components with custom hooks. Fast DOM reconciliation for real-time charts.
GIS & Mapping	Leaflet 1.9.4 + React-Leaflet 5.0.0 Leaflet.heat 0.2.0 + Turf.js 7.3.5	Hardware-accelerated GIS rendering. Dynamic marker clustering (1,000+ pins) & thermal heatmaps without UI freeze.
Visual Analytics	Recharts 3.9.2	SVG-based responsive charts with custom tooltips, theme adaptation, and time-slice animations.
Serverless Backend	Node.js LTS + Express.js 4.18.2 (Zoho Catalyst Functions)	Zero-maintenance auto-scaling serverless microservice running the unified crime_api API hub.

Primary Database	Catalyst Data Store (26 Tables) ZCQL Query Engine	Managed relational-structured storage. ZCQL queries with multi-table joins and row-level security.
AI / LLM Engine	GLM-4.7-Flash (30B MoE) Catalyst QuickML Endpoint	High-performance Mixture-of-Experts foundation model handling domain-specific criminal law tool-calling.
MFA & Cryptography	Node.js Crypto (native) AES-256-GCM + RFC 6238 TOTP	Zero-dependency cryptographic verification. Eliminates supply chain risks. Mobile Authenticator compatible.
Reporting & Export	jsPDF 2.5.2 + AutoTable 3.8.4 html2canvas 1.4.1	Client-side direct PDF generation with high-resolution canvas capture for immediate printed intelligence reports.

3. Data Architecture & Relational Schema

The Karnataka State Police FIR database contains 26 normalized tables. To achieve millisecond page loads without running hundreds of separate database queries, MADHUKAR AI utilizes a two-tier caching and schema selector pipeline.

Category	Table Name	Primary Fields	Operational Function
Core Records	CaseMaster	CaseMasterID, CrimeNo, DistrictID, PoliceStationID, CrimeMajorHeadID, CrimeMinorHeadID, BriefFacts, Lat, Lng	Master FIR registry containing every reported crime incident.
Persons Involved	Accused Victim ComplainantDetails	Name, Age, Gender, Profession, CasteID, CaseMasterID	PII-protected tables recording individuals associated with cases.
Investigation	ArrestSurrender ChargesheetDetails	AccusedID, ArrestDate, CSType (A/B/C), FilingDate	Tracks apprehension, custody, and judicial chargesheet filings.
Crime Taxonomy	CrimeHead CrimeSubHead Section Act	CrimeMajorHeadID, CrimeMinorHeadID, ActSection, OffenceSeverity	10 Major Heads + 50 Sub-heads mapped to Indian Penal Code / BNS sections.
Geo-Jurisdiction	District Unit (Police Station) Employee	DistrictID, DistrictName, UnitID, PolicePersonID, Designation	30 Karnataka districts, hundreds of police units, and officer rosters.
Security & 2FA	UserMFA	UserEmail, EncryptedSecret, IsActive, LastVerifiedAt, BackupCodes	Persistent per-officer TOTP secrets (AES-256 encrypted).

The 10 Authoritative KSP Crime Major Heads:

1. **Crimes Against Body** (Murder, Assault, Kidnapping) | 2. **Crimes Against Property** (Burglary, Dacoity, Robbery, Theft) | 3. **Crimes Against Women** (Cruelty, Dowry Death, Molestation) | 4. **Crimes Against Children** (POCSO, Child Labour) | 5. **Cyber Crimes** (UPI Fraud, Identity Theft, Hacking) | 6. **Economic Offences** (Cheating, Bank Fraud, Counterfeiting) | 7. **Narcotics** (NDPS Act, Ganja, Trafficking) | 8. **Public Order** (Rioting, Unlawful Assembly) | 9. **Arms Act** (Illegal Firearm Possession) | 10. **Crimes Against SC/ST** (POA Act Atrocities).

Data Cache Pipeline (dataCache.js):

To achieve zero-lag responses across the application, dataCache.js runs an in-memory cache in the serverless function. On initial cold start, it executes parallel ZCQL queries across all 26 tables using Promise.all(). Large datasets like CaseMaster are paged using offset cursors (300 rows/page). Once warmed, the entire operational schema is served from memory with a **5-minute TTL**. Frontend consumers request /api/app-data once on launch, hydrate the client-side store, and navigate between pages with instant 0-millisecond response times.

4. Feature-by-Feature Operational Workflows

4.1 Command Center Dashboard (Home / Operations)

Non-Technical Flow: The primary screen displayed on command center video walls. Officers monitor state-wide active cases, heinous crimes requiring immediate senior intervention, monthly crime rate trajectories, and district-by-district workload distributions.

Technical Details: Routed at / (DashboardModern.js). Consumes normalized caseViews[] from schemaSelectors.js. Integrates DateFilterContext for dynamic time windows (7D, 30D, 90D, 1Y, Custom). Renders 4 top-level KPI metric cards, a 12-month area trend chart, a 10-category donut breakdown, and an interactive Heinous Crime Action Queue. Utilizes resolveCrimeMajorHead() to ensure zero misclassifications.

4.2 GIS Spatial Crime Map & Heatmap Analysis

Non-Technical Flow: An interactive geographic map of Karnataka. Commanders can zoom from statewide views down to individual street corners, switch to a thermal heat-glow mode to spot crime clusters, and click any incident pin to inspect the FIR summary.

Technical Details: Routed at /map (CrimeMap.js). Powered by Leaflet.js with dynamic tile switching (CartoDB Dark/Light). Features **Leaflet.markercluster** for client-side spatial indexing of 1,000+ incidents and **Leaflet.heat** for Gaussian density mapping. Uses **@turf/turf** for centroid and radius calculations. Marker pins are dynamically color-coded by Crime Major Head ID.

4.3 Criminal Network Intelligence Graph

Non-Technical Flow: Reveals hidden syndicates and multi-case offender networks. When a serial burglar or gang operates across different police stations under different FIRs, this graph automatically links the cases together based on shared accomplices and identical modus operandi.

Technical Details: Routed at /network (NetworkGraph.js). Built on a high-performance HTML5 Canvas force-directed graph engine. Nodes represent Criminals (Red), Cases (Blue), Police Stations (Purple), and Victims (Green). Edge generation is powered by the custom **Forensic Similarity Engine** (similarityEngine.js). Includes a sleek **Time Range Filter Slider** and a **Similar Pattern Matches (MO) Toggle**.

4.4 Case Overview & AI Pattern Linkages

Non-Technical Flow: The core investigative dossier for individual FIRs. Displays case facts, registered legal sections, assigned investigating officers, and an AI-generated list of similar past cases that may help solve the current investigation.

Technical Details: Routed at /cases/:caseId (CaseOverview.js). Embeds mini-GIS coordinates, investigation timelines, and chargesheet milestones. Employs findSimilarCases() to compute pairwise forensic similarity scores across the entire database, surfacing top-5 matches with score percentages ($\geq 60\%$). Protects sensitive details with the PII Redaction Guard.

4.5 MADHUKAR AI Investigation Copilot

Non-Technical Flow: A conversational AI assistant tailored specifically for police officers. Officers can ask questions in natural language — such as *'Show me all robbery repeat offenders in Mysuru during August'* or *'Which police station reported the most cyber fraud?'* — and receive precise analytical answers accompanied by case tables and visual charts.

Technical Architecture & Tool-Calling Loop: Hosted at /copilot with backend in copilotEngine.js and glmClient.js. Rather than relying on simple prompt engineering, the engine implements a full **ReAct (Reasoning + Action)** tool-calling loop using Catalyst's **GLM-4.7-Flash (30B MoE)** foundation model.

Step	Actor	Operation & Technical Detail
1. Query Input	Officer (UI)	Submits natural language prompt via chat bar. Frontend posts to /api/copilot/chat.
2. Intent Parsing	copilotEngine	Scans prompt for 30 district names, 8 crime keyword groups, temporal keywords (today, month, year).
3. LLM Prompting	glmClient	Sends conversation history + system prompt + tool declarations to Catalyst QuickML endpoint.
4. Tool Calling	GLM-4.7-Flash	Model determines it needs database facts and emits a tool-call (e.g. <code>getCaseStats({district:'mysuru', crime:'robbery'})</code>).
5. Execution	dataCache	Backend executes tool against in-memory cache, formats JSON result, and feeds back to LLM.
6. Response	GLM → UI	Model synthesizes findings into professional police intelligence prose with actionable recommendations.

4.6 Predictive Crime Intelligence & Risk Scoring

Non-Technical Flow: Helps leadership deploy police patrols proactively. Evaluates historical recurrence patterns to identify which stations and districts are forecasted to experience crime surges over the next 30 days.

Technical Details: Routed at /predictions (Predictions.js). Implements a multi-factor risk scoring algorithm combining: Heinous status (+40 pts), Repeat offender history (+15 pts/incident), Unresolved investigation aging (+20 pts), and Station geographic density. Displays 30-day forecast curves with 7-day moving averages and geographical anomaly detectors.

4.7 Comprehensive Crime Statistics & Demographics

Non-Technical Flow: Analytical workspace for criminologists and senior statisticians. Explores temporal crime rhythms (what time of day and what days of the week crimes occur) alongside victim and accused demographic breakdowns.

Technical Details: Routed at /statistics (modularized into features/statistics/). Renders: 24-hour diurnal crime curve, 7-day weekly cadence bar charts, 10-head comparative volumes, Chargesheet A/B/C ratio pies, and bivariate demographic distributions (Age, Gender, Occupation, Caste).

4.8 Automated Operational Reports Generator

Non-Technical Flow: One-click generation of official, printable police reports. Officers can select daily station summaries, heinous crime briefs, or monthly district reviews and export polished PDFs ready for court or ministry presentation.

Technical Details: Routed at /reports (Reports.js). Implements client-side compilation using **jsPDF** and **jsPDF-AutoTable** with **html2canvas** for embedding visual charts. Supports 6 standard templates, multi-column data formatting, and instant CSV/JSON exports via fileExports.js.

4.9 Suspect History Timeline & Evidence Workspace

Non-Technical Flow: Provides deep criminal tracking. Displays a chronological vertical timeline of an accused person's entire criminal career across Karnataka, linking every arrest, bail status, and court hearing. The Evidence Workspace acts as a digital case diary for forensic items.

Technical Details: Routed at /suspect-timeline and /evidence-workspace. Links cross-case accused records using normalized name fuzzy-matching and KGID officer associations. Gated with PII access control.

5. Security Architecture & Zero-Trust PII 2FA

Under the Digital Personal Data Protection (DPDP) Act and police security protocols, crime records contain sensitive Personally Identifiable Information (PII) — including victim identities, minor records (POCSO), and witness contact details. MADHUKAR AI implements a strict **Zero-Trust Break-Glass Model**.

State / Role	PII Visibility	Cryptographic & Session Controls
Default State (Any Analyst / Viewer)	REDACTED Names: [REDACTED - PII PROTECTED] Contacts & addresses masked.	Read-only operational analytics. Cannot view victim/accused identities without break-glass verification.
Command Mode (Verified Officer)	UNMASKED Full legal names, addresses, and accused profiles displayed.	Requires dynamic 6-digit TOTP verification from Google/Microsoft Authenticator. Auto-locks after 2.5 minutes of inactivity.
Audit Trail	Full immutable session log.	Every unlock records: Officer KGID, Station, Timestamp, Event ID, and simulated ECDSA digital signature.

The Dynamic Per-Officer MFA Lifecycle (UserMFA Table):

Unlike basic prototypes that use a single shared password or hardcoded secret, MADHUKAR AI assigns every individual officer a unique, cryptographically generated 160-bit Base32 TOTP secret.

- First-Time Onboarding:** Officer enters Badge (KGID) and Official Email. Backend generates a unique Base32 key, encrypts it via **AES-256-GCM** with an authentication tag, stores it in Catalyst DataStore (UserMFA table), and generates a high-resolution QR code (otpauth:// URI). The officer scans this with Google Authenticator on their phone and enters their first 6-digit code to activate.
- Subsequent Unlocks:** Backend checks UserMFA. If active, it skips the QR code and prompts directly for the 6-digit code. The server decrypts the officer's secret and performs mathematical RFC 6238 HMAC-SHA1 verification.
- Self-Service Device Recovery (Lost Phone):** If an officer loses their phone, clicking 'Lost phone or cannot access Authenticator?' generates a 6-digit verification code stored in **Catalyst Cache** (5-min TTL) and delivered via

Catalyst Mail to their official email. Confirming the email code instantly revokes the compromised key and generates a fresh QR code.

6. Forensic Intelligence Engines

6.1 Forensic Modus Operandi Similarity Engine (similarityEngine.js)

Detecting crime series requires comparing criminal behavior patterns across thousands of records. The Forensic Similarity Engine evaluates 6 distinct investigative signals to produce a normalized score from 0 to 100:

Investigative Signal	Weight	Forensic Scoring Logic
Shared Accused	+30 pts	Exact or alias match between named suspects across both FIRs.
Shared Crime Major Head	+20 pts	Both cases fall under identical primary legal head (e.g. Property Crime).
Shared Sub-Head	+15 pts	Specific offence alignment (e.g. Night House Burglary vs Chain Snatching).
Modus Operandi Keywords	+15 pts	NLP keyword extraction from BriefFacts (e.g. 'iron rod', 'broken lock', 'fake UPI').
Shared Police Station	+10 pts	Hyper-local geographical correlation (same station jurisdiction).
Shared District	+10 pts	Broader jurisdictional alignment across neighboring stations.

The Cross-Category Safety Gate:

A critical vulnerability in simple recommendation systems is false linkages — for example, linking a House Burglary to a Murder simply because both occurred in the same district on the same night. The Similarity Engine enforces an authoritative **Cross-Category Gate**: If Case A and Case B belong to fundamentally distinct categories (e.g. Property vs Body, or SC/ST vs Property), they are **strictly forbidden from linking** unless a verified named accused is physically present in both cases. Links are only published if the composite score reaches **60/100 or higher**.

6.2 Crime Taxonomy & Classification Resolver (crimeTaxonomy.js)

In raw police databases, sub-head records frequently suffer from ambiguous categorization. crimeTaxonomy.js enforces strict resolution priority:

- BriefFacts Keyword Detection:** Inspects FIR text for unambiguous terms (e.g., 'burglary', 'dowry', 'ganja').
- Authoritative SUBHEAD_TO_HEAD_MAP:** 50-entry mapping binding every sub-head to its correct statutory Major Head.
- Database Lookup Name:** Uses the relational join string if valid.
- Numeric ID Map Fallback:** Maps numeric major IDs 1-10.
- General Crime Fallback:** Eliminates 'Unknown' values entirely.

7. Deployment, Hosting & Offline Resilience

MADHUKAR AI is optimized for deployment on the **Zoho Catalyst Cloud** environment. It supports both cloud deployment and dual-mode offline execution to guarantee zero presentation risk.

Deployment Target	Deployment Command	Runtime Behavior
Full Platform Deployment	<code>catalyst deploy</code>	Compiles React bundle to build/ and deploys Express microservice to Catalyst Serverless.
Frontend Only	<code>catalyst deploy --only client</code>	Uploads static SPA assets to Catalyst CDN under /app/.
Backend Only	<code>catalyst deploy --only functions</code>	Deploys functions/crime_api with all dependencies and config.
Offline Demo Mode	<code>REACT_APP_DATA_SOURCE=sample npm start</code>	Zero network dependency. Runs 1,400+ synthetic cases with 60 FPS performance without Wi-Fi.

Zero-Downtime Resilience Features:

- **In-Memory Fallback (localMfaProfiles):** If Catalyst DataStore encounters a transient network timeout or permission scope lock, the backend gracefully falls back to memory storage — allowing the officer to scan the QR code and unlock PII without 500 errors.
- **SQL Datetime Normalization:** Automatically converts ISO dates (.toISOString()) to Catalyst's strict YYYY-MM-DD HH:mm:ss format, preventing INVALID_INPUT database rejections.
- **Admin Elevation Scope:** The backend initializes Catalyst with { scope: 'admin' }, ensuring API operations execute with Project Administrator privileges regardless of client authorization tokens.

8. REST API Reference (crime_api)

Method & Route	Description & Parameters	Output Schema
GET /api/app-data	Master dataset. Returns all 26 tables in a single cached payload.	{ _meta, CaseMaster:[], District:[], ... }
GET /api/dashboard	Computes statewide totals, monthly trends, and district charts.	{ totals, monthlyTrend, districtBreakdown }
GET /api/cases	Paginated case list with filter params (districtId, crimeHeadId).	{ cases:[], limit, offset }
POST /api/copilot/chat	Conversational AI query. Body: { message, history:[] }.	{ reply, intent, toolsUsed:[] }
GET /api/mfa/status	Check officer MFA status. Params: email, badgeId.	{ status:'SETUP_REQUIRED' 'ENROLLED', qrDataUrl, secret }
POST /api/mfa/verify	Validate 6-digit Authenticator code. Body: { email, token }.	{ success:true, verified:true }
POST /api/mfa/request-reset	Lost phone flow. Dispatches email verification OTP via Catalyst Mail.	{ success:true, emailSent:true }

POST /api/mfa/confirm-reset	Confirms email OTP, revokes old secret, issues new QR code.	{ success:true, qrDataUrl, secret }
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